Fiscal Year:	FY 2006	**Task Last Updated:**	FY 01/08/2007
PI Name:	Soller, Babs R. Ph.D.		
Project Title:	Non-Invasive Sensor for Measurement of Muscle Metabolism During Exercise		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Smart Medical Systems Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	Previous affiliation was University of Massachusetts Medical School, Worcester, MA, until October 2010 (Ed., 11-9-2010)		
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI
Start Date:	07/01/2004	**End Date:**	09/30/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	2
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Hagan, R. (NASA JSC)		
Grant/Contract No.:	NCC 9-58-SM00001		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Physical exercise is the most important countermeasure currently used on the International Space Station (ISS) to mitigate muscle atrophy and bone loss, and to maintain cardiopulmonary function capacity. On ISS, assessment of health status is determined from a Periodic Fitness Evaluation (PFE) conducted starting on flight day 14 and every 30 days thereafter until the completion of the missions. This medical requirement is described in document, MR080L, Cardiovascular Physical Fitness Evaluation: Cycle Ergometry. Another medical requirement is certification for extravehicular activity (EVA). This test is conducted two weeks prior to a scheduled EVA. The test using arm cycle ergometry is described in document MR038L. In both tests, heart rate response in relation to ergometer workload is used to assess physical work capacity. However, the PFE calls for the measurement of oxygen uptake to assess aerobic capacity. Presently, the requirement for metabolism gas analysis is waived due to limitations of the equipment. However, new technologies using near infrared spectroscopy (NIRS) to determine muscle oxygenation levels may provide a simple and straightforward measure of aerobic capacity. Thus, the goal of this project is to evaluate a new NIRS monitoring device for the assessment of crew health and performance capacities. Near infrared spectroscopy (NIRS) can be used to simultaneously and continuously measure tissue oxygenation parameters, including muscle oxygen tension (PO2) and muscle pH. These parameters can be measured noninvasively on specific muscles to continually assess muscle metabolism during exercise. Previous studies indicate that oxygen consumption, which is traditionally measured with a metabolic gas analyzer, can be calculated from near infrared spectra. This project will investigate the relationship between whole-body oxygen uptake and local tissue muscle oxygenation. The findings may allow reassessment of the need for metabolic gas analysis during the periodic fitness evaluation, and the use of heart rate during arm ergometry testing for certification of EVA. The device might also provide a quantitative assessment of pre-EVA hand-grip strength (MR081L, Physical Fitness Evaluation: Handgrip Dynamometry) where no metric currently exists. The first NIRS sensor that was developed was used to study muscle oxygenation responses during exhaustive rhythmic handgrip dynamometry exercise. Preliminary evaluation of on-going data collection suggests that the exercise protocol that we are using may not be sensitive enough to detect changes muscle oxygenation patterns between endurance and strength-trained athletes
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The metabolic sensor developed as part of this project and our other NSBRI project (“Noninvasive Measurement of Blood and Tissue Chemistry”) as earth applications in critical care medicine, exercise physiology, sports medicine, and rehabilitation. This Space Medicine Project is aimed at the last 3 applications. Both aerobic fitness and muscle strength are lost during long periods of immobility, whether they be in space, or as a result of bed-confining illness. The sensors developed in this project are expected to be able to noninvasively assess loss of fitness and improvement in strength and aerobic capacity as a result of training or rehabilitation. The regular measurement of lactate threshold is a very common training technique for high performance, as well as “weekend” athletes. The noninvasive measurement of muscle pH as a surrogate for measuring lactate with a blood sample would revolutionize training by allowing continuous, noninvasive measurement throughout exercise.
Task Progress:	This year we worked on protocols for handgrip exercise and cycle ergometry. There was significant progress in both areas. Handgrip Exercise to Assess Fitness for EVA Tasks We completed one study of 10 subjects, each of whom performed 9 different handgrip protocols. The objective was to identify the protocol, which could be completed in less than 5 minutes, yet reliably tested handgrip strength. As a result of this study, we selected a test of 4 sec contraction, with 2 sec relaxation at 40% mean voluntary contraction (MVC) for further evaluation. We collected spectra during this study and found some subjects were not measured properly. The spectroscopic problem was identified and fixed. We began a second study to (1) test the reliability of the 4/2, 40% MVC protocol and (2) determine if subject response during this protocol allowed us to distinguish between trained and untrained subjects. This study was planned for 40 subjects. At this time, 18 subjects have been completed and data analysis is underway. The “trained” group was to consist of rock climbers, but it has been difficult to find rock climbers to participate in the study. Cycle Ergometry We completed a laboratory study at UMMS where we validated the new leg NIRS system against femoral venous blood. We also designed and built a new sensor holding mechanism to allow collection of spectra from the leg while cycling, without motion artifacts distorting the spectra. This was successfully tested and shipped to JSC for the leg pilot study, along with the new leg measurement system. We completed 10 subjects in a pilot cycling test where we used the NIRS system to measure muscle PO2, muscle oxygen saturation (SmO2), muscle pH, and hematocrit, along with the metabolic cart to measure oxygen uptake (VO2) and blood to measure lactate. We developed a method of using only spectroscopically measured parameters to calculate a relative measure of VO2. This measurement was well correlated with VO2 determined with the metabolic cart (R2=0.96). We were also able to show trending of blood lactate with noninvasively determined hydrogen ion concentration, potentially allowing the noninvasive measurement of lactate threshold. The NSBRI PI completed a document describing her experience in developing technology in collaboration with NASA. This document was distributed to NSBRI management.
Bibliography Type:	Description: (Last Updated: 03/22/2019)
Articles in Peer-reviewed Journals	Soller, B R, Hagan, R D, Shear, M, Walz, J M, Landry, M, Anunciacion, D, Orquiola, A and Heard, S O. "Intramuscular pH, PCO2 and PO2 during rhythmic handgrip exercise." Physiol Meas. Submitted for Publication. Sep 2006. , Sep-2006
Articles in Peer-reviewed Journals	Soyemi, O O, Soller, B R, Landry, M R, and Yang, Y. "Measurement of Tissue Oxygen Saturation Using Single-Distance, Multi-wavelength Near Infrared Spectroscopy." J Biomed Opt. Submitted for Publication. Sep 2006. , Sep-2006

Articles in Peer-reviewed Journals	Yang Y, Shoer L, Soyemi OO, Landry MR, Soller BR. "Removal of analyte-irrelevant variations in near-infrared tissue spectra." Appl Spectrosc. 2006 Sep;60(9):1070-7. PMID: 17002833 , Sep-2006
Articles in Peer-reviewed Journals	Yang, Y, Soyemi, O O, Landry, M R and Soller, B R. "Noninvasive in vivo measurement of venous blood pH during exercise using NIR Reflectance Spectroscopy." Applied Spectroscopy. In Press. Sep 2006. , Sep-2006
Patents	60/809,238. Patent Jan-2007 Soyemi, O and Soller, B. "Measuring Tissue Oxygen Saturation."
Patents	11/411,538. Patent Jan-2007 Yang, Y, Soller, B, Soyemi, O, and Shear, M. "Systems and Methods for Correcting Optical Reflectance Measurements."